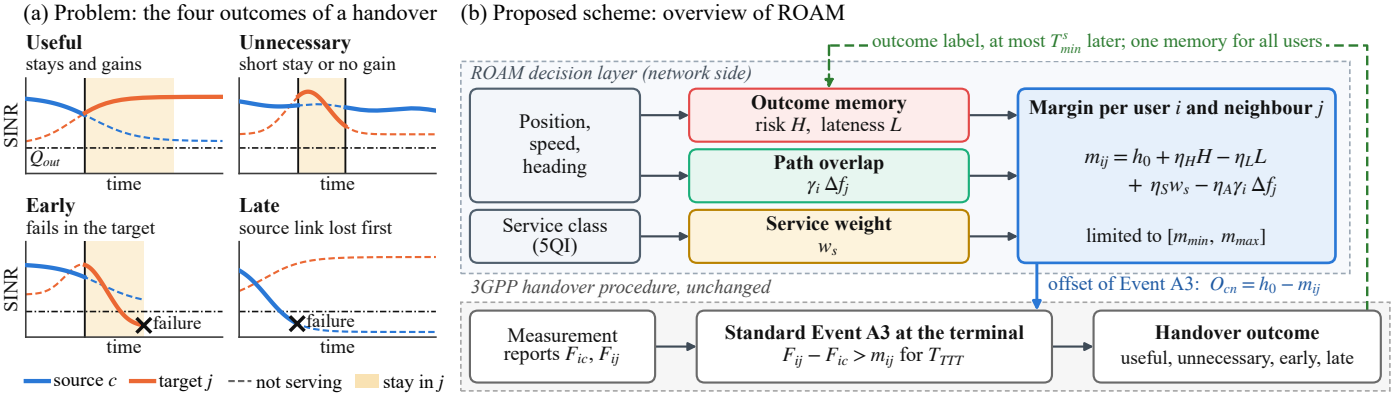

(a) Problem: the four outcomes of a handover
Useful
stays and gains
Unnecessary
short stay or no gain
SINR
Q_{out}
time
Early
fails in the target
Late
source link lost first
failure
source c
target j
not serving
stay in j
(b) Proposed scheme: overview of ROAM
outcome label, at most T^s_{min} later; one memory for all users
ROAM decision layer (network side)
Position, speed, heading
Service class (5QI)
Outcome memory
risk H, lateness L
Path overlap
$\gamma_i\,\Delta f_j$
Service weight
w_s
Margin per user i and neighbour j
$m_{ij} = h_0 + \eta_H H - \eta_L L$
$+\ \eta_S w_s - \eta_A \gamma_i\,\Delta f_j$
limited to $[m_{min},\ m_{max}]$
3GPP handover procedure, unchanged
offset of Event A3: $O_{cn} = h_0 - m_{ij}$
Measurement reports F_{ic}, F_{ij}
Standard Event A3 at the terminal
$F_{ij} - F_{ic} > m_{ij}$ for T_{TTT}
Handover outcome
useful, unnecessary, early, late